The quantitative Banach–Saks equality question

Exact negative answer in later literature

Literature-status note for `fa_banach_001`

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Status: literature already answered. Silber's Example 6.2 explicitly answers the final question of arXiv:1406.0684 negatively. This is not an original result of the run.

Original question

Bendová, Kalenda, and Spurný ask in the final question of Section 6, page 17 of arXiv:1406.0684 [1]:

> Let X be a Banach space and $A \subset X$ a bounded set. Is it necessarily true that
> $$\operatorname{wbs}(A) = 2\operatorname{sm}(A) = 2\operatorname{wus}(A)?$$

Exact later answer

Silber's arXiv:2111.12773 [2], Example 6.2 on pages 27–28, equips a Schreier space with the equivalent norm

$$\|x\|_* = \max\{\|x^+\|, \|x^-\|\}$$

and takes $A = \{e_n : n \in \mathbb{N}\}$. The example proves

$$\operatorname{sm}_1(A) = \frac{1}{2}, \qquad \operatorname{wus}_1(A) = 1, \qquad \operatorname{wbs}_0(A) = 1.$$

The paragraph immediately following the example records the notation bridge

$$\operatorname{wbs} = \operatorname{wbs}_0, \qquad \operatorname{sm} = \operatorname{sm}_1, \qquad \operatorname{wus} = \operatorname{wus}_1,$$

quotes the original equality, and explicitly states that Example 6.2 answers the question negatively. Therefore

$$\operatorname{wbs}(A) = 2\operatorname{sm}(A) = 1 \quad \text{but} \quad 2\operatorname{wus}(A) = 2.$$

This is an exact, complete, author-identified negative answer rather than an agent-inferred implication. A neighboring higher-order equality remains open in the supporting paper, but it is distinct from the 2015 question.

References

[1] H. Bendová, O. F. K. Kalenda, and J. Spurný, *Quantification of the Banach–Saks property*, J. Funct. Anal. 268 (2015), 1733–1754; arXiv:1406.0684.

[2] Z. Silber, *Quantification of Banach–Saks properties of higher orders*, arXiv:2111.12773 (2021), Example 6.2.